

[Friday lectures $\rightarrow$ L1]

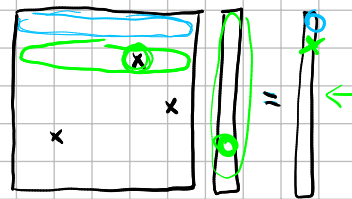
The conjugate gradient algorithm

Linear system $Ax=b$ $A \in \mathbb{R}^{n \times n}$ $b \in \mathbb{R}^n \rightarrow$ solving it via LU factorization costs $O(n^3)$.

Sparse matrices $A = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \rightarrow \text{nnz}(A) \ll n^2$

$\leadsto$ storage cost: proportional to $\text{nnz}(A)$.

$\leadsto$ matrix-vector multiplication $O(\text{nnz}(A))$



Already seen: gradient descent (GD) $\leadsto$ works for symmetric positive definite A

We will see: conjugate gradient (CG) $\leadsto$ works for SPD A

GMRES $\leadsto$ all matrices

MINRES $\leadsto$ symmetric matrices.

} later in the course...

iterative algorithms: they construct a sequence $x_0, x_1, x_2, \dots \in \mathbb{R}^n$

that (hopefully) converge to the solution $x_* = A^{-1}b$

GD: minimize $f(x) := \frac{1}{2} x^T A x - b^T x$

$\nabla f(x) = Ax - b \leadsto$ residual computed at x

input: $x_0 \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$ SPD, $b \in \mathbb{R}^n$, $K =$ number of iterations

$$r_0 = Ax_0 - b$$

for $j = 1, 2, \dots, K$

$$\alpha_j = -r_{j-1}^T r_{j-1} / (r_{j-1}^T \boxed{A r_{j-1}})$$

can do it once (per iteration) and reuse it!

$$x_j = x_{j-1} + \alpha_j r_{j-1}$$

$$r_j = Ax_j - b = \underbrace{Ax_{j-1}}_{=r_{j-1}} + \alpha_j \underbrace{A r_{j-1}}_{\boxed{A r_{j-1}}} - b = r_{j-1} + \alpha_j \boxed{A r_{j-1}}$$

end

computational cost: $O(nK + K \cdot \text{matvec})$

If the GD algorithm breaks down, it means that $r_j^T A r_{j-1} = 0$

Since A is SPD, this implies that $r_{j-1} = 0$. But $r_{j-1} = Ax_{j-1} - b = 0$

$\Rightarrow$ this means that x_{j-1} is the solution of the linear system 😊

$$\text{err}(x_j) := \frac{1}{2} (x_j - x_*)^T A (x_j - x_*)$$

$$\text{err}(x_j) \leq \text{err}(x_0) \cdot \left(\frac{\kappa - 1}{\kappa + 1} \right)^{2j}, \text{ where } \kappa = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}$$

Def: Given $A \in \mathbb{R}^{n \times n}$ (invertible), we define the CONDITION NUMBER

$$\text{as } \kappa(A) := \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} = \|A\| \cdot \|A^{-1}\|$$

$\hookrightarrow$ it's a positive number ≥ 1

and it tells you "how difficult it is to work with A ".

Def: Given an SPD $A \in \mathbb{R}^{n \times n}$, the function $\|\cdot\|_A : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$\|v\|_A = \sqrt{v^T A v} \text{ is called the "A-NORM". } \leftarrow \text{it's really a vector norm!}$$

special case: $A = I \leadsto$ Euclidean norm

$$\text{Theorem: For GD, } \|x_j - x_*\|_A \leq \|x_0 - x_*\|_A \cdot \left(\frac{\kappa(A) - 1}{\kappa(A) + 1} \right)^j$$

Idea $\leadsto$ there are other sensible ways to choose the "search directions"!

the directions for the updates of our approximations of the solution of the linear system.

step 1: $A = I$

Given $b \in \mathbb{R}^n$ we want to minimize $f(y) = \frac{1}{2} \|y - b\|^2$

$$f(y) = \sum_{i=1}^n \frac{1}{2} (y_i - b_i)^2 = f_1(y_1) + f_2(y_2) + \dots + f_n(y_n),$$

$$\text{where } f_i(y_i) = \frac{1}{2} y_i^2 - y_i b_i + \frac{1}{2} b_i^2$$

let us start with $y_0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ and we choose $e_1, e_2, \dots, e_n$ as search directions

$$y_1 \in \underset{y = y_0 + \alpha e_1}{\text{argmin}} f(y) \quad \rightarrow y_1 = \begin{bmatrix} b_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$y_2 \in \underset{y = y_1 + \alpha e_2}{\text{argmin}} f(y) \quad \rightarrow y_2 = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ 0 \end{bmatrix}, \dots, y_j = \begin{bmatrix} b_1 \\ \vdots \\ b_j \\ \vdots \\ 0 \end{bmatrix} \}_{j=1}^n$$

optimality property: at each step we have $y_j \in \underset{y \in \text{span}\{e_1, \dots, e_j\}}{\text{argmin}} f(y)$,
that is, y_j is the vector of the form $\begin{bmatrix} * \\ \vdots \\ * \\ 0 \end{bmatrix}_{n-j}$ which is closest to b .

step 2: Let's choose another orthonormal basis of $\mathbb{R}^n$

$$u_1, u_2, \dots, u_n \in \mathbb{R}^n \quad U = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \dots & u_n \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times n} \text{ orthogonal}$$

$$\text{We still want } \min f(y) = \frac{1}{2} \|y - b\|^2$$

we can write $b = u_1 c_1 + u_2 c_2 + \dots + u_n c_n$ for some $c_1, \dots, c_n \in \mathbb{R}$

$$b = U c$$

We start with $y_0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ and at the j -th step we solve

$$y_j \in \underset{x}{\operatorname{argmin}} f(y_{j-1} + x u_j) \quad (*)$$

How to do this? For any $y \in \mathbb{R}^n \exists d_1, \dots, d_n \in \mathbb{R}$ s.t. $y = U d$

$$\Rightarrow f(y) = \frac{1}{2} \|y - b\|^2 = \frac{1}{2} \|U d - U c\|^2 = \frac{1}{2} \|U(d - c)\|^2 = \frac{1}{2} \|d - c\|^2$$

$$\leadsto y_0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad y_1 = U \cdot \begin{bmatrix} c_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad y_2 = U \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ 0 \end{bmatrix}, \dots$$

optimality: $y_j \in \underset{y \in \operatorname{span}(u_1, \dots, u_j)}{\operatorname{argmin}} f(y)$, $f(y_0) \geq f(y_1) \geq f(y_2) \geq \dots \geq f(y_n) = 0$

(so: the objective function is decreasing at each iteration and the exact convergence is reached after at most n steps.)

step 3: another change of basis:

Given $R \in \mathbb{R}^{n \times n}$ invertible, consider $y = R x$ ($x = R^{-1} y$).

$$\Rightarrow \min_{y \in \mathbb{R}^n} \frac{1}{2} \|y - b\|^2 = \min_{x \in \mathbb{R}^n} \frac{1}{2} \|R x - b\|^2 = \min_{x \in \mathbb{R}^n} \left(\frac{1}{2} x^T \underbrace{R^T R}_A x - \underbrace{b^T R}_v x + \text{const} \right)$$

$$= \min_{x \in \mathbb{R}^n} \frac{1}{2} x^T A x - v^T x + \text{constant.} \quad (R^T b =: v)$$

$$\tilde{f}(x) = \frac{1}{2} \|y - b\|^2 = \frac{1}{2} \|R x - R x_*\|^2 = \frac{1}{2} (x - x_*)^T \underbrace{R^T R}_A (x - x_*) = \frac{1}{2} \|x - x_*\|_A^2$$

solution will be $x_* = R^{-1} b$

$$= R^{-1} R^T v$$

$$= (R^T R)^{-1} v$$

$$= A^{-1} v$$

$b = R^T v$

the solution is the sol. of $A x = v$

For y

Choose search directions $u_1, \dots, u_n$ orthonormal basis of $\mathbb{R}^n$

starting from $y_0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

we defined $y_j \in \underset{y \in \operatorname{span}\{u_1, \dots, u_j\}}{\operatorname{argmin}} f(y)$

For x

Choose search directions

$R^{-1} u_1, R^{-1} u_2, \dots, R^{-1} u_n$

starting from $x_0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

we find $x_j \in \underset{x \in \operatorname{span}\{d_1, \dots, d_j\}}{\operatorname{argmin}} \tilde{f}(x)$

optimality: $\tilde{f}(x_0) \geq \tilde{f}(x_1) \geq \tilde{f}(x_2) \geq \dots \geq \tilde{f}(x_n) = 0$

Def: Given a SPD $A \in \mathbb{R}^{n \times n}$, we say that $v, w \in \mathbb{R}^n$ are A-ORTHOGONAL if $v^T A w = 0$

these are A-orthogonal:

$$\text{for } i \neq j \text{ we have } d_i^T A d_j = u_i^T R^{-T} A R^{-1} u_j \\ = u_i^T \cancel{R^{-T} R^T} \cancel{R R^T} u_j = u_i^T u_j = 0$$

step 4: we make special choices for the directions and we solve min. problem exactly.

conjugate gradient (CG)

Algorithm: $A \in \mathbb{R}^{n \times n}$ SPD, $v \in \mathbb{R}^n$, $x_0 \in \mathbb{R}^n$, K = number of iterations

1. $r_0 = Ax_0 - v$
2. $d_0 = r_0$
3. for $j = 1, 2, \dots, K$
4. $\alpha_j = -r_{j-1}^T r_{j-1} / (d_{j-1}^T A d_{j-1})$
5. $x_j = x_{j-1} + \alpha_j d_{j-1}$ *
6. $r_j = r_{j-1} + \alpha_j A d_{j-1}$
7. $\beta_j = (r_j^T r_j) / (r_{j-1}^T r_{j-1})$
8. $d_j = r_j + \beta_j d_{j-1}$
9. endFor

Theorem: For CG we have $\|x_j - x_*\|_A \leq 2 \|x_0 - x_*\| \cdot \left(\frac{\sqrt{\kappa(A)} - 1}{\sqrt{\kappa(A)} + 1} \right)^j$